VitalTwin

An Explainable Personal Health Digital Twin

Happiest Health Digital Twin Challenge 2026

1. Problem

Lifestyle indicators are often viewed separately. VitalTwin creates an interactive educational view that lets a user explore a combined, transparent wellness profile and hypothetical habit changes.

2. Proposed solution

A lightweight web dashboard with user-adjustable inputs, a component-based wellness index, a synthetic trend chart, explanations of score contributions, and an intervention sandbox.

3. Healthcare use case

Health education and self-reflection. It can support a conversation about habits but is not designed for diagnosis, clinical risk prediction, or treatment decisions.

4. Technical implementation

Python, Streamlit, Pandas, NumPy, and Plotly. A deterministic, weighted rule-based engine keeps the prototype explainable. No clinical ML model is claimed because no validated dataset is included.

5. Architecture

User inputs → normalization → explainable twin engine → dashboard and component explanations. Intervention controls recalculate a hypothetical scenario using the same engine.

6. Demo plan

Show profile controls, inspect the current index and component chart, review synthetic trend, adjust activity/sleep/stress, compare the scenario, then explain limitations and privacy choices.

7. Outcomes and evaluation plan

Prototype outcomes: working interactive simulation, transparent score components, scenario comparison, tests for bounded outputs, and no external patient-data dependency. Future evaluation should assess usability and clinical appropriateness with qualified reviewers before any health claims.

8. Limitations and responsible use

All scores are illustrative and not clinically validated. Synthetic data is used for charts. Do not use the tool to change medication or treatment. Real-world deployment would require consent, security controls, governance, and appropriate clinical/legal review.

Team details: replace this line with team name, member names/roles, college, and contact. Add genuine screenshots and measured results before final submission.